

AVQACL++: Toward a Robust Framework and Benchmark for Audio-Visual Question Answering Continual Learning

Kaixuan Wu, Xinde Li, *Senior Member, IEEE*, Xinglin Li, Kezhu Zuo, and Zhiwei Lv

Abstract—This paper presents a unified framework for continual learning in Audio-Visual Question Answering (AVQACL), designed to enhance fine-grained scene understanding and spatio-temporal reasoning in dynamic multimodal task environments. To simulate realistic incremental learning scenarios, we construct two large-scale benchmark datasets, Split-AVQA and Split-MUSIC-AVQA, by reorganizing existing AVQA corpora into sequential tasks. Empirical results show that conventional models suffer from severe performance degradation and catastrophic forgetting when learning across audio, visual, and textual modalities in a continual setup. To address these challenges, we propose a novel approach that integrates four key modules: (1) Question-Guided Cross-modal Information Fusion (QCIF), which dynamically extracts task-relevant multimodal features via question-aware attention; (2) Task-specific Knowledge Distillation with Spatial-Temporal Feature Constraints (TKD-STFC), which preserves semantic output behavior and internal reasoning trajectories across tasks; (3) Question Semantic Consistency Constraint (QSCC), which regularizes evolving question representations to maintain linguistic stability. (4) Dual-Strategy Exemplar Selection (DSES), a memory-efficient replay strategy that jointly maximizes sample informativeness and diversity. All components are theoretically grounded, and formal analysis is provided to ensure modality alignment, spatial-temporal coherence, and exemplar selection reliability. Extensive experiments on both datasets demonstrate that our method consistently outperforms prior state-of-the-art AVQACL baselines in terms of accuracy, retention, and robustness.

Index Terms—Multimodal Continual Learning, Audio-Visual Question Answering, Video Scene Understanding, Spatial-Temporal Reasoning, Catastrophic Forgetting.

I. INTRODUCTION

AUDIO-VISUAL Question Answering (AVQA) is a multimodal reasoning task that requires the integration of visual and auditory cues from videos to answer natural language questions [1], [2]. Unlike traditional visual QA systems [3]–[5], AVQA involves fine-grained scene understanding, spatial-temporal localization, and auditory event recognition, which are capabilities essential for intelligent agents [6]–[11] in tasks such as human-computer interaction, robotic navigation, and context-aware multimedia retrieval.

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Despite recent advancements, most AVQA methods [12]–[15] assume static learning settings, where the full dataset is available upfront. These models lack adaptability to new tasks, suffer from catastrophic forgetting, and require expensive retraining when new data arrives. In real-world deployments, however, audio-visual data often arrives in a non-stationary, task-evolving manner. This highlights the need for continual learning paradigms that can adapt to new audio-visual concepts while preserving knowledge from previous tasks.

To address this challenge, the field of Multimodal Continual Learning (MCL) has emerged as a promising research direction [16]–[18]. MCL enables models to incrementally learn from multimodal data streams across vision, audio, and language while maintaining performance on prior tasks without access to historical data. However, AVQA under the continual learning setting remains underexplored, and there is a pressing need for benchmark datasets, challenge definitions, and unified methods tailored to this task.

To fill this gap, we introduce AVQACL [19], a benchmark and method that unify AVQA with continual learning. We construct two new datasets, Split-AVQA and Split-MUSIC-AVQA, by reorganizing AVQA [1] and MUSIC-AVQA [2] into task streams. Each task introduces new question types, auditory events, or object categories. As shown in Fig. 1, this setup mimics realistic continual learning scenarios and allows for incremental evaluation of audio-visual reasoning under memory and computation constraints.

AVQACL poses new challenges: for each task, the model must answer questions from a large candidate set using video, audio, and question inputs, while avoiding degradation on earlier tasks. This setting requires continual spatial-temporal reasoning, multimodal attention, and semantic stability across evolving knowledge distributions.

In our original conference work [19], we identified three core challenges unique to continual audio-visual question answering (AVQA), arising from the dynamic interplay between audio, visual and textual modalities over time:

- **Challenge 1: Question-Relevant Feature Extraction.** Different AVQA questions require attention to distinct spatio-temporal and modal regions. Without guided selection, irrelevant signals from past or concurrent tasks may dominate, suppressing task-specific information.
- **Challenge 2: Spatial-Temporal Reasoning Drift.** Many AVQA tasks depend on precise temporal ordering and spatial localization (e.g., “Which event happened first?”). Continual updates may disrupt learned spatial-temporal structures, degrading reasoning accuracy.
- **Challenge 3: Semantic Consistency in Question Un-**

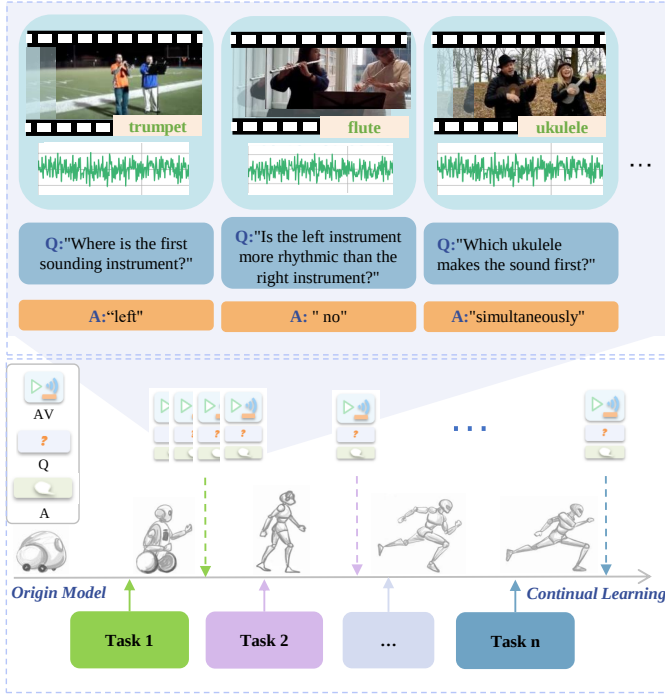


Fig. 1: Illustration of the AVQA system under a continual learning setting. The model sequentially learns across task streams where each task introduces novel question types, audio categories, or visual contexts. As the task sequence progresses, the system must retain previously acquired multimodal reasoning abilities while adapting to new concepts, without accessing past task data.

derstanding. Identical or similar questions may recur across tasks with different contextual answers. Without semantic regularization, embeddings for these questions may drift, harming generalization and attention alignment.

While our conference framework addressed the challenges above, it assumed a naive memory replay mechanism for prior knowledge retention. In this journal version, we identify a critical additional challenge specific to resource-constrained continual learning environments:

- **Challenge 4: Memory-Efficient Knowledge Retention.** Under realistic memory constraints, naive exemplar sampling may cause unbalanced coverage, overfitting frequent samples or omitting rare cases. This motivates the need for a principled exemplar selection strategy that jointly considers informativeness and diversity to preserve representative knowledge within a fixed replay buffer.

To address these challenges, we propose a modular continual learning framework with four key components:

- **Question-Guided Cross-modal Information Fusion (QCIF)** addresses Challenge 1 by using question-aware attention to filter audio-visual features, ensuring semantic alignment and compact relevance.
- **Task-specific Knowledge Distillation with Spatial-Temporal Feature Constraints (TKD-STFC)** tackles Chal-

lenge 2 by preserving intermediate feature dynamics and spatial reasoning pathways across tasks.

- **Question Semantic Consistency Constraint (QSCC)** solves Challenge 3 by enforcing stability in the question embedding space via cosine regularization.
- **Dual-Strategy Exemplar Selection (DSES)** addresses Challenge 4 and balances representativeness and diversity to optimize replay buffer efficiency with theoretical coverage guarantees.

We evaluate our method on Split-AVQA and Split-MUSIC-AVQA, comparing it with six strong continual learning baselines [20]–[25]. Our model consistently outperforms prior work in accuracy, forgetting mitigation, and cross-modal reasoning robustness.

Compared to our previous conference version [19], this journal extension offers four substantial enhancements:

- 1) We propose the Dual-Strategy Exemplar Selection (DSES) module to improve memory efficiency via diversity-aware replay, with formal analysis of coverage and retention guarantees.
- 2) We provide theoretical justification for all modules, including attention-guided fusion, spatio-temporal preservation, and semantic regularization.
- 3) We incorporate Transformer-based modality encoders to assess generalization and verify the framework's modularity across backbone choices.
- 4) We conduct extensive new experiments with deeper ablation studies, qualitative visualizations, and robustness evaluation under realistic AVQACL settings.

II. RELATED WORKS

A. Audio-Visual Question Answering

Audio-Visual Question Answering [26]–[28] has emerged as a pivotal task in multimodal AI, enabling models to reason over dynamic visual content and corresponding audio signals to answer natural language queries. Initial studies focused on audio-aware extensions of visual QA datasets and models [29], [30]. These approaches laid the foundation for integrating audio cues into temporal visual understanding.

With the introduction of purpose-built datasets such as AVQA [1] and MUSIC-AVQA [2], the field has expanded toward domain-specific reasoning. AVQA incorporates real-world object-sound interactions, while MUSIC-AVQA emphasizes fine-grained auditory-visual associations in musical contexts. These benchmarks have catalyzed the development of structured attention frameworks and semantic-aware feature fusion [13], [31], which model spatial, temporal, and semantic dependencies across modalities.

Recent advances explore richer supervision signals (e.g., contrastive, question-guided attention) and flexible answering formats to enable open-ended reasoning [12], [32], [33]. Models like PACS [14] enhance generalization by learning inter-modal reasoning patterns under weak supervision. Additionally, real-world applications have inspired robust cross-modal alignment techniques for noisy environments [34].

B. Continual Learning on Unimodal Tasks

Continual Learning (CL) [35]–[37] has been extensively studied in unimodal domains such as image classification, with the goal of mitigating catastrophic forgetting when new tasks are sequentially introduced. Key solution strategies fall into three major categories: (1) Parameter Regularization. Methods like EWC [25], MAS [24], and LwF [21] estimate the importance of each model parameter for previously learned tasks. These importance weights guide selective regularization, limiting parameter drift for crucial neurons during updates. (2) Knowledge Distillation. This line of work leverages output or feature alignment between old and new models [38]–[40]. PODNet [38], for instance, aligns intermediate representations, while other approaches minimize output divergence to preserve decision boundaries. (3) Memory-Based Replay. Methods such as iCaRL [22] and Mnemonics [41] store a fixed subset of samples from previous tasks. These exemplars are replayed during training to counteract forgetting. Variants like GSS [42] and DER [43] refine the selection strategy based on gradient contribution or knowledge uncertainty. While these methods achieve notable success in unimodal scenarios, they typically lack the flexibility to handle multimodal interactions or hierarchical semantic structures found in AVQA.

C. Multimodal Continual Learning

Extending continual learning to multimodal tasks introduces new challenges, including alignment drift between modalities, modality-specific forgetting, and asynchronous feature evolution. Multimodal Continual Learning (MCL) [44] thus requires simultaneous retention of cross-modal representations and task-specific knowledge.

Recent efforts in MCL explore structural constraints and regularization strategies. For instance, AV-CIL [20] introduces cross-modal similarity and visual attention distillation to maintain inter-modality coherence during task transitions. The Class-Incremental Generalization Network (CIGN) [45] leverages dedicated modality-specific tokens and relational distillation to stabilize learning. HAD [16] further enriches intra-task consistency by enforcing hierarchical feature retention across temporal segments. Memory-based multimodal methods [46] also emerge as promising solutions. CavRL [47] introduces contrastive distillation across audio and visual domains, complemented by exemplar memory buffers. Meanwhile, large-scale video-based CL benchmarks such as vCLIMB [17] and PIVOT [48] incorporate temporal consistency and modality-aware prompts to evaluate realistic video reasoning under continual learning.

A unique direction is pursued by VQACL [18], which introduces a dual-granularity formulation for question type and concept stream expansion, mirroring human-like curriculum acquisition in audio-visual tasks. Our work is inspired by this trend, and we further generalize the continual AVQA setting to open-ended, fine-grained answering spaces with spatial-temporal feature preservation and semantic consistency constraints.

III. PROBLEM DEFINITION OF AVQA CONTINUAL LEARNING

Let us define a sequence of tasks in the continual learning framework as $\mathcal{T} = \{\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_n\}$, each corresponding to a different subset of the audio-visual question answering (AVQA) dataset. The objective is to optimize a unified model $\mathcal{F}_\theta$, parameterized by θ , that incrementally acquires multi-modal reasoning capabilities from inputs comprising audio x^a , video x^v , and natural language questions x^q , and outputs the correct answer y at every step of task progression.

For the t^{th} incremental learning task $\mathcal{T}_t$, the associated training set is denoted by:

$$D_t = \{(x_{t,i}^a, x_{t,i}^v, x_{t,i}^q, y_{t,i})\}_{i=1}^{n_t} \quad (1)$$

where each tuple includes the audio signal $x_{t,i}^a$, visual sequence $x_{t,i}^v$, and corresponding textual question $x_{t,i}^q$ for the i^{th} sample, with $y_{t,i} \in \mathcal{A}_t$ representing the corresponding answer drawn from the label space $\mathcal{A}_t$ of task $\mathcal{T}_t$.

Following prior continual learning settings [20], we maintain a memory buffer of limited capacity, denoted by M_t , which stores a fixed number of exemplars from previously learned tasks. At each step, the model is trained on a combined dataset:

$$D_t^m = D_t \cup M_t \quad (2)$$

The learning objective at incremental step t is therefore formulated as:

$$\theta_t = \arg \min_{\theta} \mathbb{E}_{(x^a, x^v, x^q, y) \sim D_t^m} [\mathcal{L}(\mathcal{F}_\theta(x^a, x^v, x^q), y)] \quad (3)$$

where $\mathcal{L}$ denotes the task loss function, typically the cross-entropy loss over predicted answers. Upon completion of training on task $\mathcal{T}_t$, the model is evaluated on a cumulative test set containing data from all tasks $\mathcal{T}_1$ through $\mathcal{T}_t$.

IV. PROPOSED METHOD

A. Framework Overview

We introduce AVQACL++, a unified continual learning framework for Audio-Visual Question Answering (AVQA), designed to incrementally acquire, consolidate, and preserve multimodal knowledge across evolving tasks. As presented in Fig. 2, AVQACL++ features six core modules: Multimodal Feature Encoders for perceptual feature extraction, Question-Guided Cross-modal Information Fusion for aligning audio-visual features, Task-specific Knowledge Distillation with Spatial-Temporal Feature Constraints to mitigate forgetting, Question Semantic Consistency Constraint to enhance linguistic generalization, and Dual-Strategy Exemplar Selection for efficient sample selection. This modular design allows AVQACL++ to achieve continual learning with minimal forgetting and improved generalization.

B. Multimodal Representation Encoding

Given a training task $\mathcal{T}_t$, each input comprises synchronized audio A , visual frames V , and a natural language question Q . We extract modality-specific features using two encoder settings: a CNN-RNN baseline and a transformer-based variant.

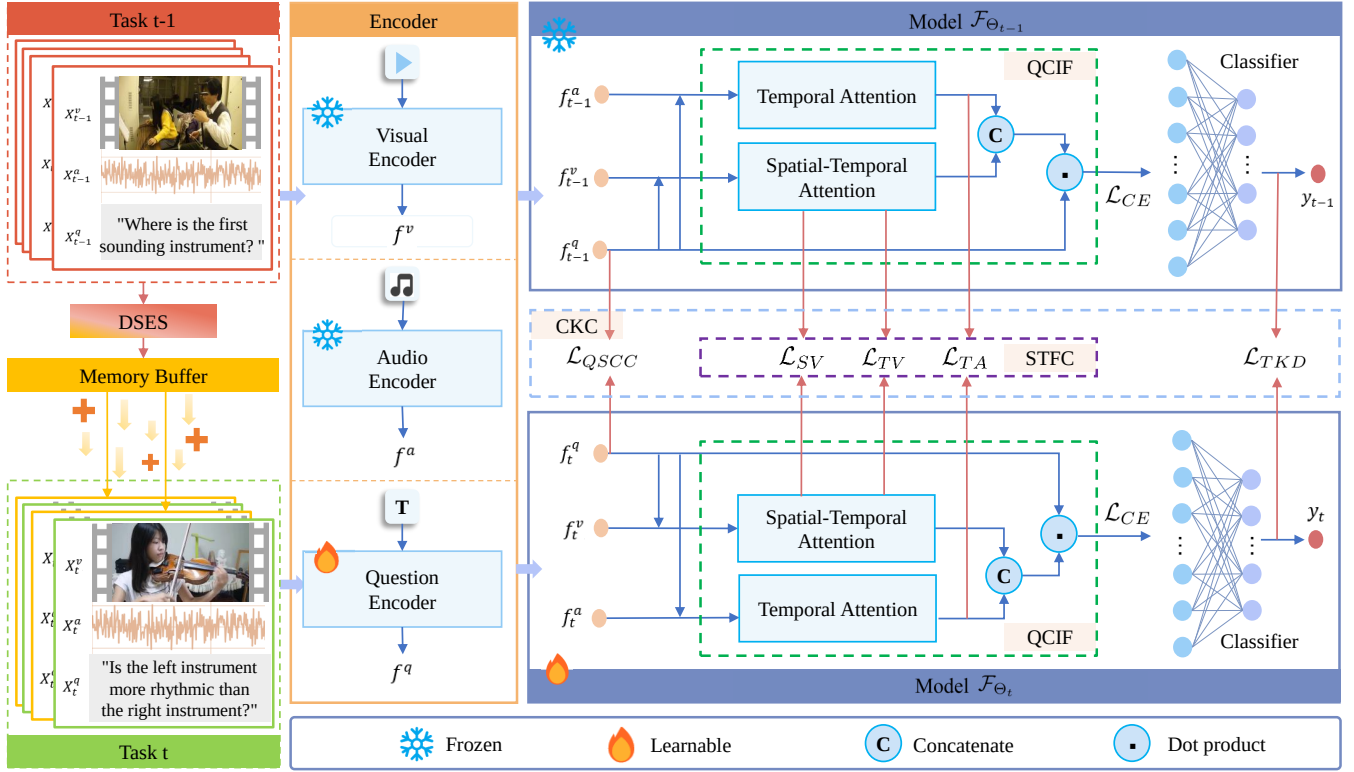


Fig. 2: Overall framework of AVQACL++. Given multimodal inputs (V, A, Q) from current task $\mathcal{T}_t$, the model selectively replays exemplars from the previous task $\mathcal{T}_{t-1}$ via the newly introduced Dual-Strategy Exemplar Selection (DSES) module. Visual and audio inputs are encoded by frozen pre-trained encoders, while the question is processed by a trainable encoder. The Question-guided Cross-modal Information Fusion (QCIF) module performs attention-based fusion aligned with the question embedding. To prevent forgetting, AVQACL++ applies two distillation losses: (1) TKD-STFC, task-specific knowledge distillation with spatial-temporal feature constraints ($\mathcal{L}_{TKD}$ and $\mathcal{L}_{STFC}$); and (2) QSCC, semantic consistency constraint on question embeddings ($\mathcal{L}_{QSCC}$). The model is trained incrementally with cross-entropy loss $\mathcal{L}_{CE}$ on current predictions while preserving past knowledge.

Audio Encoding. For each segment A_t , the baseline uses VGGish [49] to extract log-mel spectrogram features:

$$f_a^t = \text{VGGish}(A_t), \quad A_t \in \mathbb{R}^{T_a \times d_a} \quad (4)$$

The transformer-based variant uses Wav2Vec2.0 [50] on raw audio:

$$f_a^t = \text{Wav2Vec2.0}(A_t), \quad A_t \in \mathbb{R}^{L \times 1} \quad (5)$$

Visual Encoding. The video V_t is uniformly sampled into T_v frames. The baseline applies ResNet-18 [51]:

$$f_v^t = \text{ResNet}(V_t), \quad V_t \in \mathbb{R}^{T_v \times F \times d_v} \quad (6)$$

The transformer variant uses ViT [52], encoding each frame as patch tokens:

$$f_v^t = \text{ViT}(V_t), \quad f_v^t \in \mathbb{R}^{T_v \times d_v} \quad (7)$$

Question Encoding. The question Q is encoded into a global semantic vector f_q . The baseline uses a single-layer LSTM [2]:

$$f_q = \text{LSTM}(Q), \quad f_q \in \mathbb{R}^{d_q} \quad (8)$$

The transformer variant uses BERT [53], extracting the [CLS] token:

$$f_q = \text{BERT}(Q), \quad f_q \in \mathbb{R}^{d_q} \quad (9)$$

C. Question-Guided Cross-Modal Information Fusion (QCIF)

To dynamically filter irrelevant background information and align audio-visual features with the question intent, we propose a Question-guided Cross-modal Information Fusion (QCIF) module with multi-stage attention mechanisms.

1) *Visual Spatial-Temporal Attention:* Let $V \in \mathbb{R}^{T_v \times F \times d_v}$ be the visual feature sequence. Attention is computed using the question vector f_q :

$$V_{\text{att}} = \text{MultiHead}(f_q, V, V) \quad (10)$$

$$V_{\text{out}} = \text{LN}(V_{\text{att}} + \text{Dropout}(\text{ReLU}(W_v V_{\text{att}}))) \quad (11)$$

Where V refers to the projected video-level temporal feature sequence constructed from frame-wise embeddings extracted in Eq. 6 and Eq. 7. Specifically, spatial features f_v^t are obtained per sampled frame using the 2D backbone, then temporally stacked and mapped via the learnable operator $\text{Proj}(\cdot)$ to form $\mathbf{F}_v \in \mathbb{R}^{T_v \times d}$.

2) *Audio Temporal Attention:* Similarly, audio tokens $A \in \mathbb{R}^{T_a \times d_a}$ are aligned with the question:

$$A_{\text{att}} = \text{MultiHead}(f_q, A, A) \quad (12)$$

$$A_{\text{out}} = \text{LN}(A_{\text{att}} + \text{Dropout}(\text{ReLU}(W_a A_{\text{att}}))) \quad (13)$$

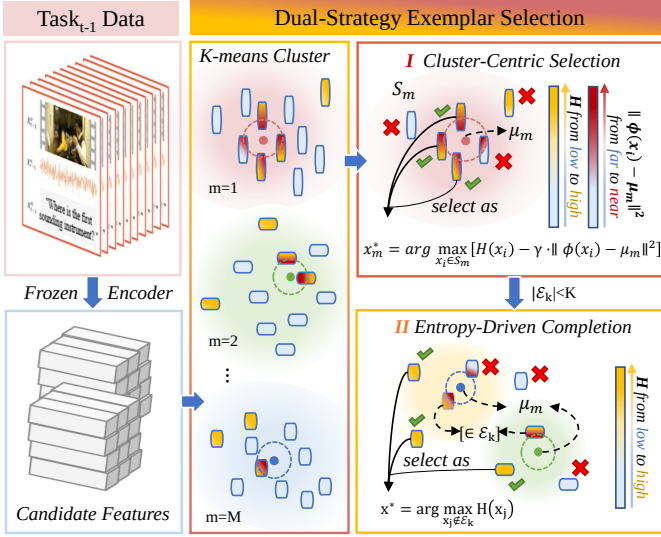


Fig. 3: Overview of the Dual-Strategy Exemplar Selection (DSES) framework. Given candidate features from previous tasks, Phase I performs cluster-centric selection via K-means to enforce spatial diversity and selects entropy-informed exemplars closest to each cluster center. Phase II compensates the underfilled memory buffer by selecting globally high-entropy samples to maximize task informativeness. This two-phase strategy ensures that the selected exemplars are both representative and diverse, effectively preserving prior task knowledge under constrained memory.

3) *Fusion*: The fused representation F is obtained by:

$$F_{av} = \text{Tanh}([A_{out}; V_{out}]) \quad (14)$$

$$f_{fused} = \text{Tanh}(F_{av} \odot f_q) \quad (15)$$

This fused vector serves as the final multimodal input to the classifier.

Intuition and Information-Theoretic Viewpoint: QCIF functions as a selective attention mechanism that extracts semantically relevant audio-visual features conditioned on the question representation Q . Inspired by the Information Bottleneck principle [54], we aim to preserve only task-useful information while filtering modality-specific noise:

$$\max I(f; Y) - \beta I(f; X_V, X_A) \quad (16)$$

Here, attention provides a differentiable approximation of this bottleneck, guiding the fusion of only question-relevant audio-visual cues. Further theoretical details and a formal proof of representational sufficiency under conditional independence assumptions are provided in the Appendix A of Supplementary Material.

D. Task-Specific Knowledge Distillation with Spatial-Temporal Feature Constraints (TKD-STFC)

Conventional knowledge distillation approaches [55] focus solely on output distributions. However, such methods are insufficient to preserve fine-grained temporal and spatial representations. To address this, we introduce Spatial-Temporal Feature Constraints (STFC) into the distillation process.

Let f^{new} and f^{old} denote the feature maps from the current and previous models, respectively. We define the first-order difference of features along spatial or temporal dimension $d \in \{t, s\}$ as:

$$\Delta_d f_i = f_{i+\delta_d} - f_i \quad (17)$$

where δ_d is the unit offset along dimension d , and $i \in \mathcal{I}_d$ is a valid index.

The feature alignment loss is then given by:

$$\mathcal{L}_d = \frac{1}{|\mathcal{I}_d|} \sum_{i \in \mathcal{I}_d} (\Delta_d f_i^{new} - \Delta_d f_i^{old})^2 \quad (18)$$

We define the total spatial-temporal constraint loss as:

$$\mathcal{L}_{STFC} = \lambda_{STFC} (\mathcal{L}_{TA} + \mathcal{L}_{TV} + \mathcal{L}_{SV}) \quad (19)$$

where $\mathcal{L}_{TA}$, $\mathcal{L}_{TV}$, and $\mathcal{L}_{SV}$ denote temporal audio, temporal visual, and spatial visual losses, respectively.

We additionally introduce a distillation loss that aligns the output distributions of the old and new models for previously seen tasks. Let z_{old} and z_{new} denote the pre-softmax logits from the old and current models, respectively. Applying temperature scaling τ yields softened probability distributions:

$$p_{old} = \text{Softmax}(z_{old}/\tau), \quad p_{new} = \text{Softmax}(z_{new}/\tau) \quad (20)$$

The task-specific knowledge distillation loss is then defined using Kullback-Leibler divergence [56]:

$$\mathcal{L}_{TKD} = \tau^2 \cdot \text{KL}(p_{old} \parallel p_{new}) \quad (21)$$

This encourages the new model to preserve task-relevant output behavior, while the STFC term constrains internal feature dynamics. The combined objective is:

$$\mathcal{L}_{TKD-STFC} = \mathcal{L}_{TKD} + \mathcal{L}_{STFC} \quad (22)$$

Further theoretical analysis is provided in the Appendix B of Supplementary Material.

E. Question Semantic Consistency Constraint (QSCC)

To avoid semantic drift in question representations across tasks, we introduce the QSCC loss that aligns question features temporally. Let q_{new}^i and q_{old}^i denote the embeddings of the i^{th} question from the current and previous models, respectively. The cosine-similarity-based loss is defined as:

$$\mathcal{L}_{QSCC} = 1 - \frac{1}{N} \sum_{i=1}^N \cos(q_{new}^i, q_{old}^i) \quad (23)$$

We integrate this constraint into the classification objective as:

$$\mathcal{L}_{CE-QSCC} = \mathcal{L}_{CE} + \lambda_{QSCC} \cdot \mathcal{L}_{QSCC} \quad (24)$$

Further semantic stability analysis is provided in the Appendix C of Supplementary Material.

F. The Overall Objective

The overall objective of our proposed framework is to integrate and retain multimodal knowledge across sequential tasks, while mitigating catastrophic forgetting and preserving task-specific features. After introducing the three key modules, QCIF for question-guided fusion, TKD-STFC for spatio-temporal knowledge distillation, and QSCC for semantic consistency, the total loss can be concisely formulated as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{TKD-STFC}} + \mathcal{L}_{\text{CE-QSCC}} \quad (25)$$

where $\mathcal{L}_{\text{TKD-STFC}}$ preserves both output behavior and spatio-temporal feature dynamics across tasks, and $\mathcal{L}_{\text{CE-QSCC}}$ ensures accurate classification while maintaining semantic stability of question embeddings. This unified loss function establishes the foundation for continual learning in our AVQA setting.

G. Dual-Strategy Exemplar Selection (DSES)

We propose a Dual-Strategy Exemplar Selection (DSES) approach that integrates both uncertainty-aware and diversity-aware sampling into a unified selection objective, which is depicted in Fig. 3. Let $\phi : X \rightarrow \mathbb{R}^d$ denote a frozen feature extractor (e.g., the multimodal fusion layer), and let θ_t, θ_{t-1} be the model parameters before and after task t , respectively. We define the exemplar utility objective as:

$$\max_{\mathcal{E}_k \subset \mathcal{D}_{t-1}} \underbrace{I(\theta_t; \theta_{t-1} \mid \mathcal{E}_k)}_{\text{Task Informativeness}} + \gamma \cdot \underbrace{\mathbb{E}_{x_i, x_j \in \mathcal{E}_k} \|\phi(x_i) - \phi(x_j)\|^2}_{\text{Feature Diversity}} \quad (26)$$

where $I(\theta_t; \theta_{t-1} \mid \mathcal{E}_k)$ denotes the task-relevant information preserved by exemplars, and $\gamma > 0$ balances the contribution of diversity.

Execution Logic. To make the selection process clearer, we summarize the execution of DSES in two stages:

- **Phase I: Cluster-Centric Selection.** We first partition the feature embeddings into M clusters. In each cluster S_m , we select one exemplar x_m^* that maximizes uncertainty while remaining close to the centroid μ_m :

$$x_m^* = \arg \max_{x_i \in S_m} [H(x_i) - \gamma \cdot \|\phi(x_i) - \mu_m\|^2] \quad (27)$$

Here, $H(x_i)$ is the predictive entropy, and γ controls the trade-off between informativeness and representativeness.

- **Phase II: Entropy-Driven Completion.** If the cluster-based selection does not fill the exemplar budget K , we greedily add remaining high-entropy samples:

$$\mathcal{E}_k \leftarrow \mathcal{E}_k \cup \left\{ \arg \max_{x_j \notin \mathcal{E}_k} H(x_j) \right\} \quad (28)$$

This ensures that all budgeted slots are used while maximizing the informativeness of exemplars.

Illustrative Example. Suppose class c has 100 samples and an exemplar budget $K = 10$.

- 1) Phase I: Use K-means to divide the 100 samples into $M = 6$ clusters. From each cluster, select one exemplar using Eq. 27, yielding 6 exemplars.

Algorithm 1 DSES: Dual-Strategy Exemplar Selection

Require: Dataset $\mathcal{D}_{t-1}$, feature extractor ϕ , exemplar count K , clusters M , weight γ

- 1: **for** each class k in previous task **do**
- 2: $\mathcal{D}_k \leftarrow \{x_i \in \mathcal{D}_{t-1} \mid y_i = k\}$
- 3: Extract features: $\Phi_k \leftarrow \{\phi(x_i) \mid x_i \in \mathcal{D}_k\}$
- 4: $[S_1, \dots, S_M], [\mu_1, \dots, \mu_M] \leftarrow \text{KMEANS}(\Phi_k, M)$
- 5: Initialize exemplar set: $\mathcal{E}_k \leftarrow \emptyset$
 - ▷ Phase I: Select one exemplar per cluster
- 6: **for** each cluster S_m **do**
- 7: $x_m^* \leftarrow \arg \max_{x_i \in S_m} [H(x_i) - \gamma \cdot \|\phi(x_i) - \mu_m\|^2]$
- 8: $\mathcal{E}_k \leftarrow \mathcal{E}_k \cup \{x_m^*\}$
 - ▷ Phase II: Fill remaining budget with high-entropy samples
- 9: **while** $|\mathcal{E}_k| < K$ **do**
- 10: $x^* \leftarrow \arg \max_{x_j \in \mathcal{D}_k \setminus \mathcal{E}_k} H(x_j)$
- 11: $\mathcal{E}_k \leftarrow \mathcal{E}_k \cup \{x^*\}$
- 12: **return** Exemplar sets $\{\mathcal{E}_k\}$ for all classes k

- 2) Phase II: Since only 6 exemplars are selected but $K = 10$, we further choose 4 additional high-uncertainty samples from the remaining pool.
- 3) The final exemplar set contains 10 samples that are both diverse (spread across clusters) and informative (high entropy).

In summary, DSES first guarantees *diversity* by enforcing cluster coverage (Phase I), and then enhances *informativeness* by filling the remaining budget with high-uncertainty samples (Phase II). This two-stage procedure ensures that the exemplar buffer preserves both representative and challenging samples, thereby improving continual generalization. Further theoretical analysis and approximation guarantees are provided in the Appendix D of Supplementary Material.

Complexity Analysis. Let $N_k = |\mathcal{D}_k|$ denote the number of candidate samples for class k , d the feature dimension, and M the number of clusters.

- Feature Extraction: $\mathcal{O}(N_k \cdot d)$ for computing $\phi(x)$.
- K-means Clustering: $\mathcal{O}(M \cdot N_k \cdot d \cdot T)$ where T is the number of clustering iterations.
- Entropy Computation: Assuming C classes, softmax + entropy costs $\mathcal{O}(C)$ per sample: total $\mathcal{O}(N_k \cdot C)$.
- Cluster-Centric Selection: For each cluster, computing $H(x_i)$ and distance to centroid is $\mathcal{O}(|S_m| \cdot d)$. Total over M clusters is $\mathcal{O}(N_k \cdot d)$.
- Entropy-Driven Completion: In the worst case, up to $K - M$ high-entropy selections from the remaining pool: $\mathcal{O}(N_k \cdot \log N_k)$ if using priority queue.

The total per-class complexity is:

$$\mathcal{O}(N_k \cdot d \cdot T + N_k \cdot C + N_k \cdot \log N_k) \quad (29)$$

Assuming $T, C \ll N_k$, this scales quasi-linearly with the number of candidates and linearly with feature dimension.

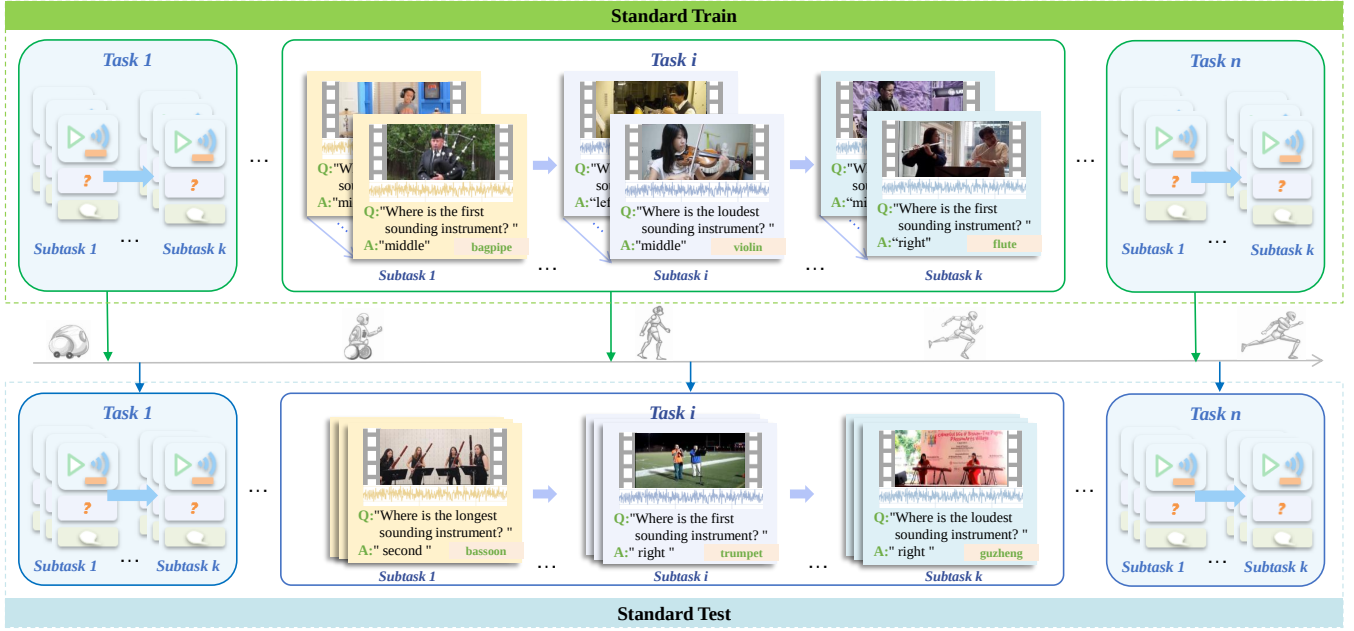


Fig. 4: The proposed AVQACL setting, which is divided into two main phases: Standard Train and Standard Test. In the Standard Train phase, multiple tasks are organized, where each task consists of sequential subtasks designed to simulate realistic incremental learning scenarios. Each subtask presents a distinct audiovisual question, guiding the model to learn in an organized manner. The Standard Test phase evaluates the model's performance across similar tasks, ensuring a comprehensive assessment of its capabilities

V. EXPERIMENTS

A. Datasets and Task Configuration

Datasets. To facilitate systematic benchmarking, we construct two continual learning datasets based on existing AVQA corpora: AVQA [1] and MUSIC-AVQA [2], both of which explore multi-sensory question answering involving spatial and temporal reasoning. We create two incremental learning benchmarks, named Split-AVQA and Split-MUSIC-AVQA, to simulate real-world continual learning scenarios. Split-AVQA comprises 39,314 video clips (10 seconds each) and 41,008 question-answer pairs, distributed across four question types: "Where," "Which," "Happening," and "Come From", along with 120 unique sound categories. Split-MUSIC-AVQA contains 7,324 60-second musical performance videos and 20,774 QA pairs. These span five question types: "Counting," "Location," "Comparative," "Temporal," and "Existential", using 19 question templates. Answer prediction is framed as an open-ended classification over 40 distinct musical entities or concepts, including instruments and ensemble attributes.

Task Construction. The AVQACL task structure is built to mirror human-like cognitive learning in multimedia contexts, where learners encounter new concepts while retaining prior knowledge. Inspired by curriculum-based approaches such as [18], we implement a hierarchical task construction scheme, as illustrated in Fig. 4. First, the full dataset is segmented into T tasks based on question type. Next, each task is further divided into K subtasks based on category groupings (e.g., object categories or sound classes). Let C be the total number of categories, evenly split into M groups for subtask partitioning. The specific configurations are:

- Split-AVQA: $T = 4, K = 6, C = 120, M = 20$
- Split-MUSIC-AVQA: $T = 5, K = 11, C = 22, M = 2$

This hierarchical design enables incremental exposure to new question types and concepts, promoting the development of robust multimodal reasoning capabilities.

B. Evaluation Metrics

To quantitatively evaluate continual learning performance, we adopt two widely accepted metrics [20]: Mean Accuracy (MA) and Average Forgetting (AF). MA measures the average performance across all tasks seen up to step T :

$$MA = \frac{1}{T} \sum_{t=1}^T a_t \quad (30)$$

where a_t is the testing accuracy after completing training on task $\mathcal{T}_t$. AF quantifies performance degradation on prior tasks due to the acquisition of new knowledge. It is defined as:

$$AF = \frac{1}{T-1} \sum_{t=2}^T \left(\frac{1}{t-1} \sum_{i=1}^{t-1} \left[\max_{\gamma < t} (a_{\gamma,i} - a_{t,i}) \right] \right) \quad (31)$$

where $a_{\gamma,i}$ denotes the model's accuracy on task $\mathcal{T}_i$ after training on task $\mathcal{T}_\gamma$, and $a_{t,i}$ is the accuracy on the same task following training on $\mathcal{T}_t$. These metrics jointly capture the model's ability to maintain prior knowledge and adapt to novel tasks, which are central goals in continual AVQA learning.

C. Implementation Details

All experiments are conducted using the PyTorch framework [57] on a single NVIDIA RTX 5500 Laptop GPU with 16

TABLE I: Performance comparison of AVQACL++ and State-of-the-Art continual learning methods on Split-AVQA and Split-MUSIC-AVQA benchmarks under two encoder settings. Mean Accuracy (MA) and Average Forgetting (AF) are reported. Buffer sizes are indicated for replay-based methods.

Method	Split-AVQA					Split-MUSIC-AVQA				
	Buffer	Encoder α		Encoder β		Buffer	Encoder α		Encoder β	
		MA $\uparrow$	AF $\downarrow$	MA $\uparrow$	AF $\downarrow$		MA $\uparrow$	AF $\downarrow$	MA $\uparrow$	AF $\downarrow$
Vanilla	None	18.34	40.94	19.83	40.45	None	28.79	33.17	28.97	31.10
LwF [21]	None	21.53	30.49	24.12	31.16	None	30.93	30.14	31.49	28.60
EWC [25]	None	18.89	40.69	21.29	38.42	None	28.76	32.58	28.57	31.62
MAS [24]	None	24.47	8.91	17.62	27.85	None	29.33	31.15	30.76	29.35
EASE [59]	None	30.46	3.04	27.82	2.78	None	31.28	28.34	30.38	29.67
DIKI [60]	None	28.39	3.73	29.17	2.58	None	30.69	29.33	28.95	29.85
C-CLIP [61]	None	31.87	2.66	28.49	2.69	None	32.38	27.89	30.75	29.15
iCaRL [22]	5000	25.28	33.76	27.33	30.03	700	28.46	33.16	29.61	30.60
SSIL [23]	5000	30.50	2.68	29.50	2.96	700	32.16	28.05	29.35	29.99
AV-CIL [20]	5000	27.49	2.91	29.83	2.69	700	31.22	29.65	28.42	29.58
HAD [16]	5000	31.54	2.78	28.74	2.83	700	32.85	27.57	30.23	29.24
AVQACL(ours) [19]	5000	32.05	2.47	27.53	2.26	700	33.64	27.08	28.44	31.16
AVQACL++ (ours)	5000	34.22	2.06	30.69	2.22	700	34.11	26.58	31.61	28.85
Upper Bound	-	55.42	-	59.84	-	-	65.57	-	61.86	-

GB memory. Unless otherwise specified, all models, including our AVQACL++ and baseline methods, are trained under identical configurations with shared initialization, optimization settings, and memory budgets. To preserve modality-specific priors, the audio and visual encoders are frozen during training, while the question encoder and fusion modules remain trainable. We use the Adam optimizer [58] with a learning rate of 1×10^{-3} and weight decay of 1×10^{-4} . Each task is trained for 120 epochs. The batch size is set to 128 for Split-AVQA and 64 for Split-MUSIC-AVQA. For both datasets, we uniformly sample 10 video frames per second. The task sequences are as follows: Split-AVQA includes Come From $\rightarrow$ Happening $\rightarrow$ Where $\rightarrow$ Which, and Split-MUSIC-AVQA consists of Counting $\rightarrow$ Existential $\rightarrow$ Location $\rightarrow$ Comparative $\rightarrow$ Temporal. The exemplar memory buffer is fixed at 5000 for Split-AVQA and 700 for Split-MUSIC-AVQA.

We tune all hyperparameters for each encoder-dataset combination. For Split-AVQA, under encoder α , we set $\lambda_{\text{STFC}} = 0.2$, $\lambda_{\text{QSCC}} = 0.2$, $\gamma = 0.5$, and temperature $\tau = 1.0$; under encoder β , we set $\lambda_{\text{STFC}} = 0.2$, $\lambda_{\text{QSCC}} = 0.6$, $\gamma = 0.6$, and $\tau = 2.0$. For Split-MUSIC-AVQA, under encoder α , we use $\lambda_{\text{STFC}} = 0.2$, $\lambda_{\text{QSCC}} = 1.0$, $\gamma = 0.5$, and $\tau = 4.0$; under encoder β , we set $\lambda_{\text{STFC}} = 1.0$, $\lambda_{\text{QSCC}} = 0.8$, $\gamma = 0.5$, and $\tau = 2.0$. All experiments are conducted using a fixed random seed to ensure reproducibility.

D. Comparison and Analysis with Baselines

Baselines. To comprehensively evaluate the effectiveness of our proposed method in the context of AVQACL, we compare it against a suite of representative and widely adopted continual learning baselines. These include:

- Vanilla: A naive baseline that incrementally trains the model on each task without any forgetting mitigation

mechanism. It serves as a reference point to measure the severity of catastrophic forgetting.

- LwF [21]: Learning without Forgetting, which preserves past knowledge by distilling outputs from the previous model while training on new tasks.
- EWC [25]: Elastic Weight Consolidation uses Fisher information to identify important parameters and penalizes changes to them during new task learning.
- MAS [24]: Memory Aware Synapses computes importance weights in a data-driven manner based on output sensitivity, providing a lightweight and task-adaptive regularization.
- EASE [59]: Instead of relying on exemplars or heavily modifying pre-trained parameters, EASE introduces lightweight task-specific adapter subspaces. These subspaces are expanded incrementally and aggregated into a high-dimensional ensemble, while a semantic-guided prototype complement strategy synthesizes old-class features without using exemplars.
- DIKI [60]: It retains pre-trained knowledge of large VLMs (e.g., CLIP) while integrating new task information. By adopting a residual knowledge integration mechanism and distribution-aware calibration, DIKI alleviates both backward forgetting and forward forgetting (loss of zero-shot ability), outperforming prompt-tuning baselines with minimal trainable parameters.
- C-CLIP [61]: C-CLIP introduces a benchmark for multi-modal continual learning and proposes contrastive knowledge consolidation combined with parameter-efficient low-rank adaptation. This approach not only mitigates catastrophic forgetting of zero-shot capabilities but also enhances downstream task performance, achieving a favorable balance between adaptability and knowledge retention.

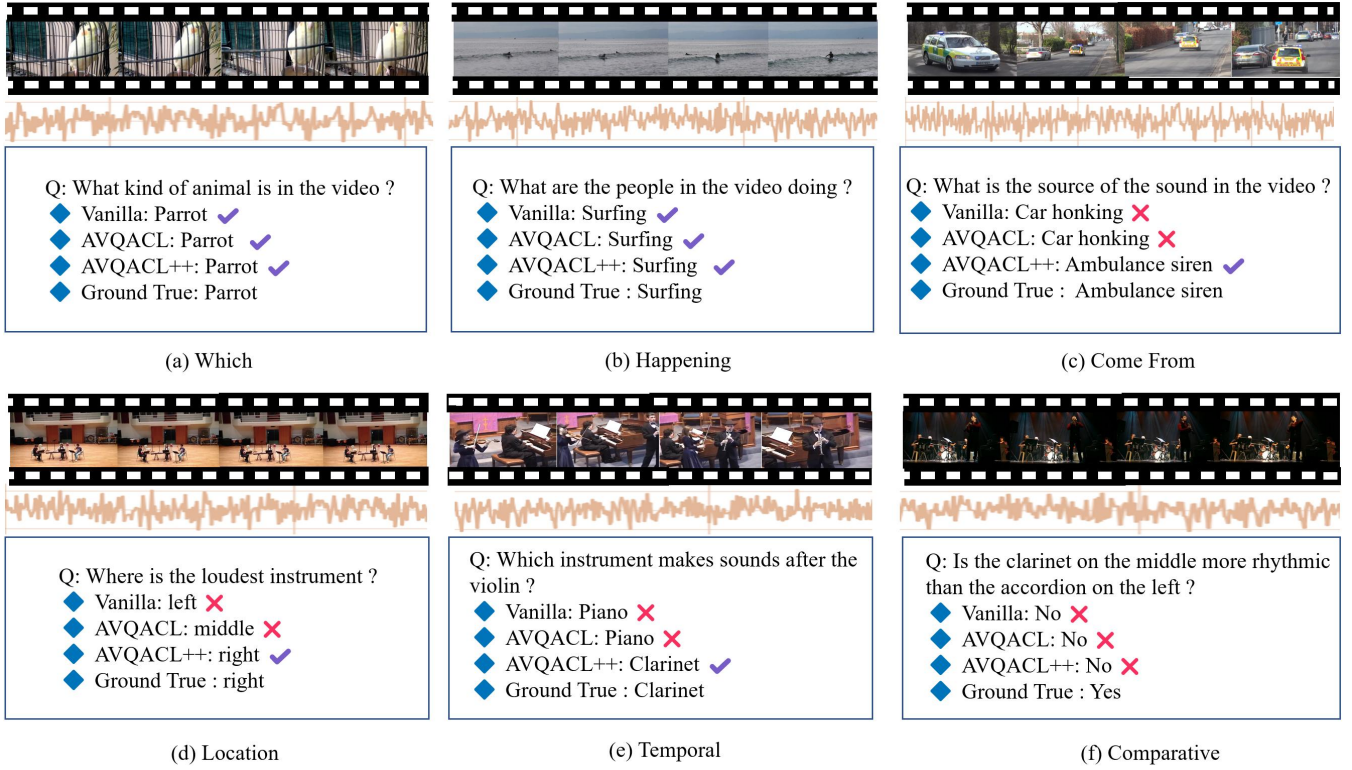


Fig. 5: Qualitative Visual Results. We compare predictions from the baseline Vanilla, AVQACL, and our proposed AVQACL++ models on representative examples from Split-AVQA and Split-MUSIC-AVQA. AVQACL++ demonstrates superior performance in most categories (a–e), accurately aligning audio-visual cues with question semantics and handling complex spatial-temporal reasoning. Notably, in the *Comparative* case (f), all models fail, revealing challenges in modeling subjective acoustic relationships. These examples highlight both the strengths and remaining limitations of AVQACL++.

- iCaRL [22]: Incremental Classifier and Representation Learning maintains a memory buffer of exemplars and updates a nearest-mean classifier, combining rehearsal and representation learning.
- SS-IL [23]: Self-Supervised Incremental Learning leverages cross-task self-supervision and contrastive learning objectives to enhance representation stability across tasks.
- AV-CIL [20]: An audio-visual continual learning method that incorporates cross-modal similarity constraints and visual attention distillation to preserve multimodal consistency.
- HAD [16]: It exploits the inherent hierarchical structures in both data (snippet-level, video-level, distribution-level) and models (low-level and high-level semantics). The method incorporates a Hierarchical Augmentation Module (HAM) and a Hierarchical Distillation Module (HDM), enabling effective preservation of both data-level and model-level knowledge, thereby reducing forgetting in audio-visual tasks.

We also report the performance of an *Upper Bound* model trained with access to all past and current data (joint learning). This serves as a performance ceiling for evaluating continual learning gaps.

Comparison and Analysis. Tab. I presents the performance comparison on Split-AVQA and Split-MUSIC-AVQA benchmarks under two encoder settings. The following observations

can be drawn:

- Baseline methods. Regularization-based methods such as LwF, EWC, and MAS provide marginal improvements over the vanilla model but fail to effectively mitigate catastrophic forgetting in multimodal reasoning. Replay-based methods such as iCaRL and SSIL achieve higher MA, but their AF values remain relatively high, reflecting insufficient stability.
- Recent CIL methods. Approaches such as EASE, DIKI, and C-CLIP perform well in vision-only tasks but transfer poorly to AVQA continual settings. This highlights that directly adapting image-centric methods to multimodal reasoning is inadequate.
- Our baseline (AVQACL). With Encoder α (ResNet-18), AVQACL achieves competitive MA (32.05) and significantly lower AF (2.47) compared to AV-CIL (27.49 MA, 2.91 AF). With Encoder β (ViT), AVQACL attains lower MA (27.53) but still improves AF (2.26). This discrepancy can be explained by the trade-off between stability and plasticity: although ViT has higher theoretical representational capacity, it is more sensitive to noisy video frames and small per-task data in continual learning, which limits its MA. However, the stronger regularization in AVQACL helps reduce forgetting, yielding better AF. Compared with AV-CIL, AVQACL thus favors stability even at the cost of slight accuracy drop.

TABLE II: Ablation study on the effect of each core component in AVQACL++. We report Mean Accuracy (MA) and Average Forgetting (AF) on both Split-AVQA and Split-MUSIC-AVQA datasets under encoder α and β settings.

Method	Split-AVQA				Split-MUSIC-AVQA			
	Encoder α		Encoder β		Encoder α		Encoder β	
	MA $\uparrow$	AF $\downarrow$	MA $\uparrow$	AF $\downarrow$	MA $\uparrow$	AF $\downarrow$	MA $\uparrow$	AF $\downarrow$
Vanilla	18.34	40.94	19.83	40.45	28.79	33.17	28.97	31.10
w/o QCIF	31.95	2.54	29.79	2.54	28.91	32.25	29.39	30.46
w/o TKD-STFC	28.38	3.87	26.63	3.43	28.75	32.13	28.54	31.46
w/o QSCC	33.63	2.32	29.47	2.94	29.62	31.33	28.64	30.56
AVQACL(w/o DSES)	32.05	2.47	27.53	2.26	33.64	27.08	28.44	31.16
AVQACL++(ours)	34.22	2.06	30.69	2.22	34.11	26.58	31.61	28.85

- Our full framework (AVQACL++) and other multimodal continual learning methods. Compared with AV-CIL and HAD, both of which are tailored for multimodal continual learning, our AVQACL++ consistently achieves the best overall results. For example, on Split-AVQA with Encoder α , AVQACL++ reaches 34.22 MA and 2.06 AF, surpassing HAD (31.54 MA, 2.78 AF). On Split-MUSIC-AVQA, AVQACL++ also outperforms HAD (34.11 MA vs. 32.85, and 26.58 AF vs. 27.57). These results show that AVQACL++ improves both multimodal reasoning accuracy and knowledge retention simultaneously, achieving a superior stability-plasticity balance.

In summary, although MA and AF sometimes exhibit different trends due to the inherent trade-off between learning new tasks (plasticity) and retaining old ones (stability), AVQACL++ consistently achieves the most favorable trade-off across benchmarks, demonstrating both high accuracy and strong resistance to forgetting. In addition, a more detailed comparison of fine-grained Mean Accuracy (MA) results across different continual learning methods is provided in Appendix E of the Supplementary Material.

E. Qualitative Visual Results and Discussion

Fig. 5 provides a set of representative examples illustrating the performance of Vanilla, AVQACL, and our proposed AVQACL++ across six AVQA question types: *Which*, *Happening*, *Come From*, *Location*, *Temporal*, and *Comparative*. These visualizations reflect both the model's strengths and its current limitations.

Overall, AVQACL++ exhibits strong multimodal reasoning capabilities. For example, in the *Come From* (c) and *Location* (d) examples, AVQACL++ is the only model to correctly identify the source of the ambulance siren and locate the loudest instrument, demonstrating improved temporal reasoning (e.g., capturing event ordering and audio onset) and spatial reasoning (e.g., localizing sound sources) enabled by its TKD-STFC and question-guided fusion design. Similarly, in *Temporal* reasoning (e), it correctly infers event ordering, which other models fail to capture.

Despite these strengths, failure cases still occur, most notably in the *Comparative* example (f), where all models incorrectly judge rhythmic differences between instruments.

This suggests a limitation in modeling fine-grained auditory comparisons or interpreting subjective semantic cues.

General Insights. These results highlight that AVQACL++ effectively aligns audio and visual signals with question semantics and maintains feature stability across tasks. However, complex comparative or subjective reasoning remains an open challenge. Future work could explore integrating metric learning or external rhythmic priors to improve reasoning about nuanced acoustic attributes.

F. Ablation study

Effect of Each Component. We conduct ablation studies on both Split-AVQA and Split-MUSIC-AVQA under encoder configurations (α , β) to evaluate the contribution of each component in AVQACL++ (Tab. II). Removing QCIF results in a notable MA drop (e.g., 34.22 $\rightarrow$ 31.95 on Split-AVQA α), highlighting its role in aligning audio-visual features with question semantics. Excluding TKD-STFC significantly increases forgetting (e.g., AF rises from 2.06 to 3.87), underscoring its importance in maintaining spatial-temporal reasoning across tasks. Disabling QSCC leads to a 0.6–1.0 point drop in MA and higher forgetting, indicating its effectiveness in preserving semantic consistency in question representations. Finally, incorporating DSES yields consistent gains over AVQACL across all settings, validating the advantage of jointly optimizing exemplar informativeness and diversity. Together, these results demonstrate that each module addresses a distinct challenge in continual AVQA, and their integration ensures robust, generalizable performance across tasks.

Sensitive of Hyperparameter. We perform a grid search over the regularization coefficients λ_{STFC} and λ_{QSCC} to analyze their impact on Mean Accuracy (MA) across different datasets and encoder configurations. Results (Fig. 6) show that moderate STFC regularization (e.g., $\lambda_{\text{STFC}} = 0.2$) combined with moderate to high QSCC (e.g., $\lambda_{\text{QSCC}} = 0.6$) yields the best performance on Split-AVQA for both CNN- and Transformer-based encoders. For Split-MUSIC-AVQA, lower QSCC values work better for Encoder α due to greater semantic variability in audio, while Encoder β benefits from stronger regularization (both $\lambda = 1.0$), likely due to its richer modeling capacity. Overall, the analysis confirms that STFC helps preserve spatial-temporal coherence, while QSCC maintains question-level

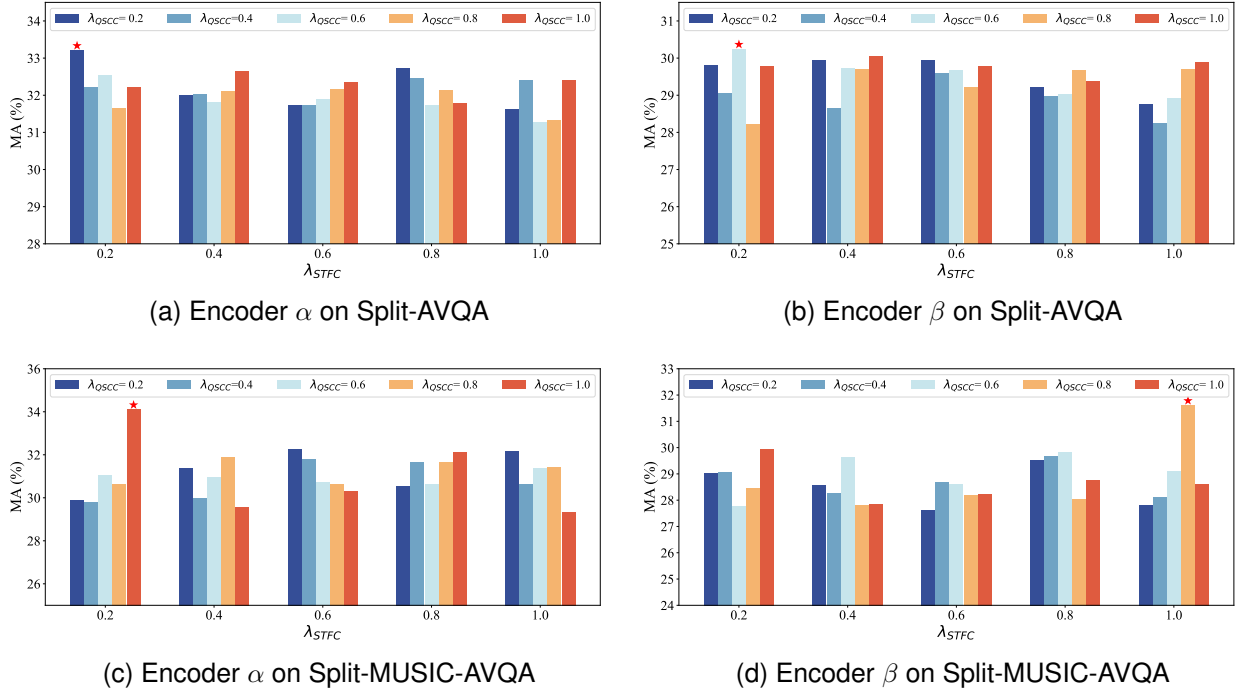


Fig. 6: Sensitivity analysis of hyperparameters λ_{STFC} and λ_{QSCC} on Mean Accuracy (MA) for Split-AVQA and Split-MUSIC-AVQA datasets under encoder α and β settings.

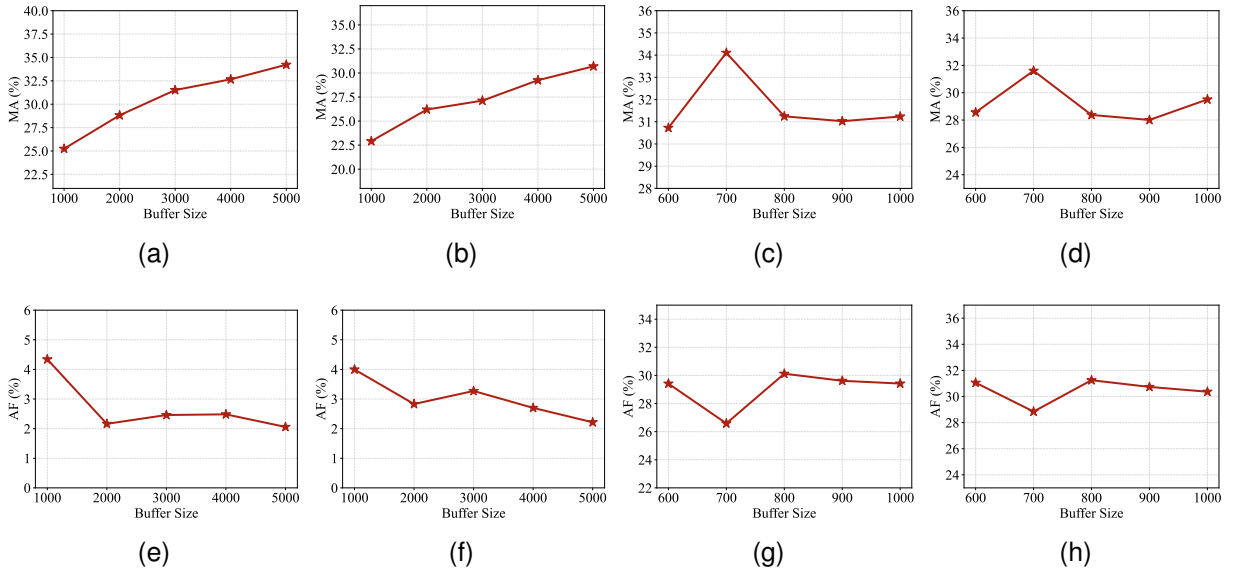


Fig. 7: The impact of buffer size. For Split-AVQA, (a) and (e) under encoder α , (b) and (f) under encoder β ; For Split-MUSIC-AVQA, (c) and (g) under encoder α , (d) and (h) under encoder β .

semantic consistency, and both should be tuned adaptively to dataset and encoder characteristics.

Impact of Buffer Size. As illustrated in Fig. 7, this experiment results demonstrate that increasing the memory buffer size positively impacts Mean Accuracy (MA), especially for Encoder α , where MA improves significantly from 25% to 35% as the buffer increases from 1000 to 5000. Encoder β shows more stable gains with slight fluctuations. In contrast,

Average Forgetting (AF) remains relatively stable across buffer sizes for both encoders, indicating that larger buffers mainly improve accuracy without substantially affecting feature retention. These results highlight the importance of buffer size tuning in continual learning for accuracy gains, while its impact on forgetting is less pronounced.

Effect of Task Order. As depicted in Fig. 8, this study highlights the critical influence of task order on continual learning

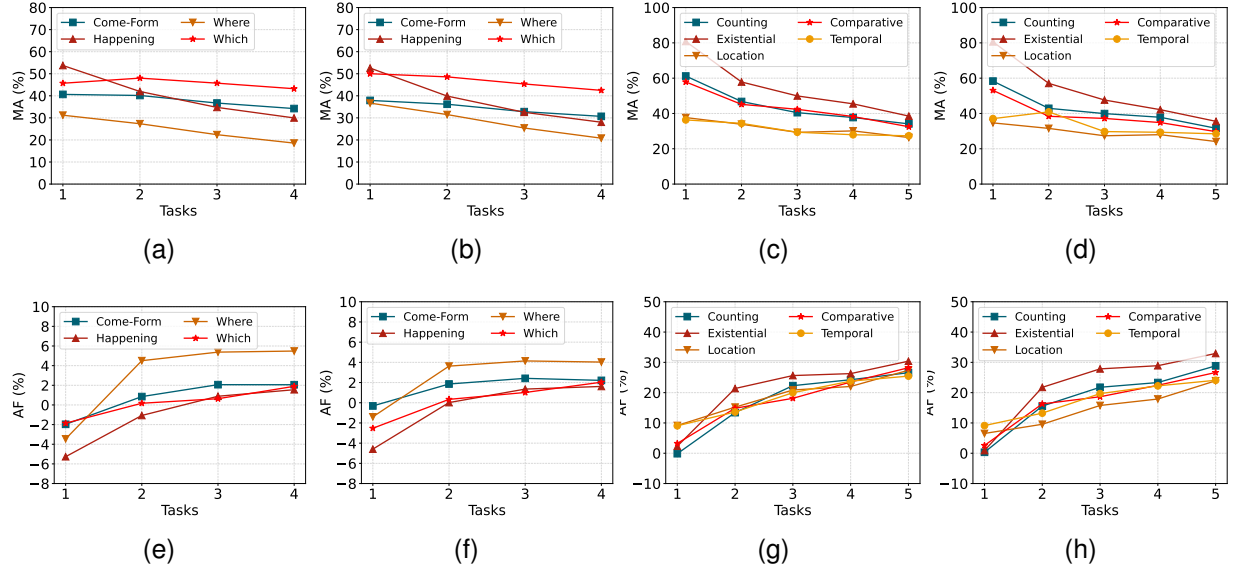


Fig. 8: The effect of task order. For Split-AVQA, (a) and (e) under encoder α , (b) and (f) under encoder β ; For Split-MUSIC-AVQA, (c) and (g) under encoder α , (d) and (h) under encoder β .

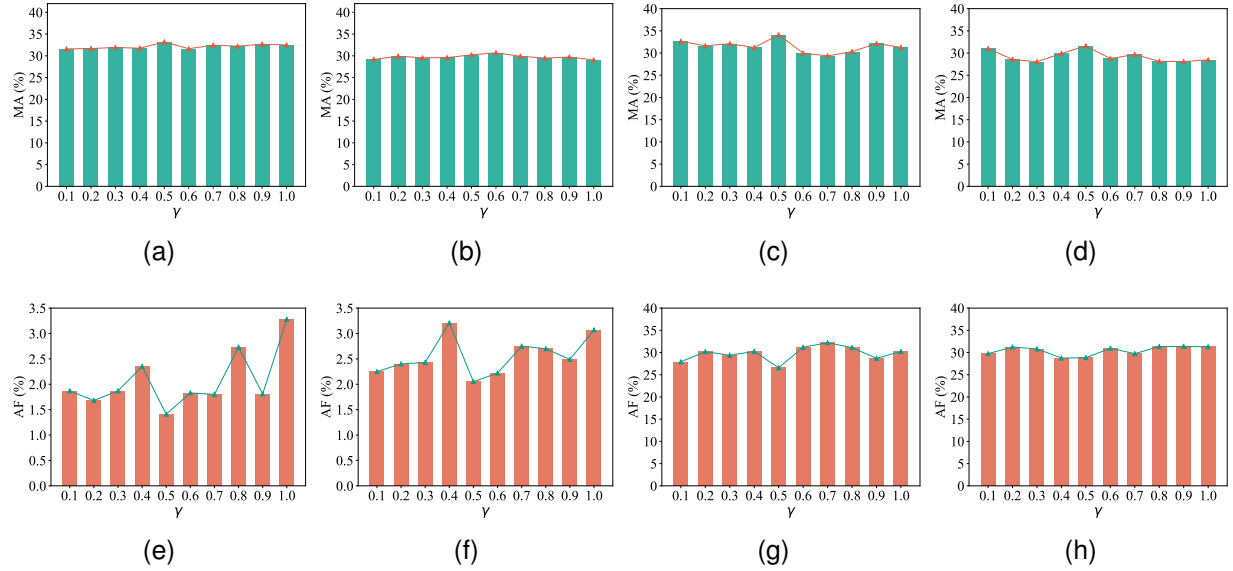


Fig. 9: The impact of varying weight γ . For Split-AVQA, (a) and (e) under encoder α , (b) and (f) under encoder β ; For Split-MUSIC-AVQA, (c) and (g) under encoder α , (d) and (h) under encoder β .

performance. Mean Accuracy (MA) tends to decline for later or more complex tasks (e.g., “Which,” “Temporal”), especially under encoder α , indicating sensitivity to task sequencing. In contrast, earlier tasks like “Come From” and “Counting” consistently achieve higher MA. Meanwhile, Average Forgetting (AF) generally increases or stabilizes across tasks, showing modest improvements in feature extraction even as accuracy drops. These findings underscore the need to carefully design task sequences to balance learning efficiency and retention in AVQA continual learning frameworks.

Impact of Varying Weight γ . We analyze the sensitivity of the Dual-Strategy Exemplar Selection (DSES) module to the

weight parameter γ , which balances predictive uncertainty and feature-space diversity. As presented in Fig. 9, experiments across Split-AVQA and Split-MUSIC-AVQA using both encoder configurations (α , β) show that Mean Accuracy (MA) remains stable across different γ values, with peak performance typically observed near $\gamma = 1.0$. Average Forgetting (AF) also shows minimal variation, with slightly lower values at $\gamma = 0.2$ – 0.4 in some cases. Overall, the results indicate that DSES is robust to the choice of γ . While fine-tuning may offer marginal improvements, the model maintains strong performance across a broad range, validating its practical applicability without requiring sensitive hyperparameter ad-

justment.

Additional ablation studies, including the impact of varying the temperature parameter τ and the effect of the number of frames per video, are presented in Appendix F of the Supplementary Material.

VI. CONCLUSION

In this paper, we present a unified framework for Audio-Visual Question Answering under continual learning settings, addressing the challenge of incremental multimodal reasoning over evolving video streams. To support systematic benchmarking, we construct two large-scale datasets, Split-AVQA and Split-MUSIC-AVQA, by reorganizing existing AVQA corpora into sequential tasks. These benchmarks simulate real-world learning dynamics and offer a comprehensive testbed for studying audio-visual continual learning. Our proposed method introduces three key modules: a Question-Guided Cross-modal Information Fusion (QCIF) mechanism for question-relevant feature alignment, a Task-specific Knowledge Distillation strategy with Spatial-Temporal Feature Constraints (TKD-STFC) for preserving reasoning capabilities, and a Question Semantic Consistency Constraint (QSCC) to maintain semantic stability across tasks. In addition, a Dual-Strategy Exemplar Selection (DSES) algorithm is incorporated to enhance memory efficiency, with theoretical approximation guarantees. All components are theoretically grounded, and extensive experiments validate the effectiveness and generalizability of our framework. Our method consistently outperforms existing continual learning baselines in terms of average accuracy, average forgetting mitigation, and multimodal reasoning fidelity, establishing a new state-of-the-art in AVQACL.

LIMITATIONS

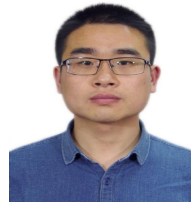
While our method demonstrates strong performance under the continual audio-visual question answering (AVQACL) setting, one limitation lies in the relatively lightweight QA model and modular encoder architecture. Our design aims to strike a balance between performance, interpretability, and computational cost, particularly important in the continual learning context. With the emergence of powerful multimodal large language models (MLLMs) such as Flamingo [62], GPT-4V [63], and LLaVA [64], a natural question is whether our continual learning framework can be extended to work with such models. Although promising, direct integration of MLLMs [65] poses several open challenges, including high memory/computation requirements, limited support for task-adaptive memory updates, and lack of native audio-stream processing. We leave this as a future research direction toward scaling continual learning for general-purpose multimodal reasoning.

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